Unbrowse Maintenance Network

Trust, Accountability, and Optional Bonding in a Shared Route Graph

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Abstract

A shared graph of reusable web routes solves discovery: an agent looks up a route someone already mapped instead of re-deriving it from a browser on every call, and the companion paper [1] shows this wins whenever the route fee undercuts the cost of rediscovery. Discovery, though, is a one-time cost; freshness is a standing liability. Routes decay — schemas drift, auth flows mutate, fields are renamed behind cheerful `200`s — and a route that is wrong with confidence is worse than no route at all. This paper takes up the maintenance problem: who keeps the routes fresh, and how is the claim that a route is fresh made trustworthy. We frame the graph as a common-pool resource that degrades under uncoordinated use [4], show why pure usage fees structurally under-provide freshness (a non-rival public good [5] whose decay is a negative externality [6]), and propose collateralised accountability as the corrective: a maintainer bonds the network's security asset against a freshness claim, and a false or conflicting claim is slashable [8], finalised by a threshold quorum [9] and adjudicated against a tamper-evident, independently auditable log [11]. We are explicit about the token this requires — why it exists strictly for the security function, why its fair distribution is a security parameter rather than a marketing choice, and why a coordination asset launched for money cannot become trustworthy — and equally explicit about the limit we have not closed, formal Sybil resistance [7]. Throughout we tag what is [shipped] shipped versus [proposed] proposed, and we stop exactly where the capture engine begins [3]: this paper is the trust surface in full, not the moat.

1 Introduction: the maintenance problem, not the discovery problem

The first problem an agent has on the open web is *discovery*: finding the route in the first place is expensive, and re-deriving it on every call is wasteful. That problem is, by now, settled. The earlier work in this series [1] reduces it to a single inequality — when the fee for reusing a route someone already mapped, $f_{\text{shared route}}$, is less than the cost of rediscovering it from scratch, $c_{\text{rediscovery}}$, the rational move is to share. A live route graph already does this: an intent resolves to a ranked endpoint shortlist and executes, and the cost of mapping each route is amortised across every agent that follows. [shipped] We are not here to relitigate that. It works.

This paper is about the *second* problem, which only shows up once the first is solved. A graph with real traffic is a graph with real decay. Schemas drift; authentication flows mutate; a field is renamed, a pagination cursor changes shape, an endpoint is deprecated behind a redirect that returns a cheerful `200`. None of this is adversarial — it is simply what live systems do. The inequality that justified sharing the route says nothing about who keeps it true a month later. A route that was correct when it was mapped and is wrong when it is used is worse than no route at all, because it is wrong with confidence: the agent does not know to fall back. Discovery is a one-time cost; freshness is a standing liability.

So the question this paper takes up is narrow and unglamorous: *who keeps the routes fresh, and how is their claim that a route is fresh made trustworthy?* Maintenance is not a feature bolted onto discovery; it is the thing that decides whether the shared graph degrades into a liability or compounds into an asset. We separate, throughout, what is running from what is designed. The route graph itself — resolution, execution, the signed record of which routes were used and what they returned — is live. [shipped] The accountability machinery that makes a freshness claim costly to fake — the bonding, challenge, and slashing layer described in the later sections — is a design, and we mark it as such. [proposed] We would rather under-claim a running system than over-claim a proposed one; the tags are there so the reader is never in doubt which they are reading.

2 The route as a common-pool resource

The shared route graph is not a public good in the textbook sense, and naming it correctly matters for the governance that follows. It is a *common-pool resource* (CPR) in the precise sense Ostrom gave the term [4]: a resource from which it is hard to exclude users, but whose value is depletable through uncoordinated use. The depletion here is not consumption of a finite stock — a route does not get used up when an agent calls it. The depletion is *epistemic*. Every agent that leans on a stale route and propagates its output as correct, every contributor who adds a route and never returns to confirm it still resolves, lowers the average trustworthiness of the graph for everyone. The classic tragedy — each participant rationally drawing down a shared resource because the cost of their own neglect is socialised — reappears exactly, with staleness in the place of overgrazing.

The reason this framing is useful, rather than merely a tidy metaphor, is that the CPR literature does not stop at diagnosing the tragedy. Ostrom's central finding is that durable commons are neither privatised nor centrally administered; they are governed by the appropriators themselves under a small set of recurring design principles [4]. Three of those principles map directly onto the layers this paper builds. *Clearly defined boundaries* — who is entitled to claim a route is fresh, and who may draw on it — become the trust-tier and identity machinery: a freshness claim is not anonymous, it is attached to an accountable party. *Graduated sanctions* — penalties that scale with the severity and repetition of a violation rather than a single all-or-nothing punishment — become the slashing tiers, where a first low-confidence miss is treated differently from a repeated or high-value false claim. *Nested enterprises* — governance organised in layers, smaller units inside larger ones — become the trust tiers themselves: open routes governed lightly for low-risk traffic, sitting inside more tightly governed high-trust routes for authenticated or high-value paths.

We borrow the template, not the institution. Ostrom studied irrigation systems, fisheries, and forests — human appropriators with faces and long memories. A route graph traversed by agents has neither, which removes some of the informal social enforcement her commons relied on and forces the corresponding work onto explicit, mechanical accountability. The economic case for why that explicit machinery is necessary at all — why pure usage fees under-provide freshness, and why a bond is the corrective rather than an embellishment — is taken up next; the cryptographic and settlement primitives that make the sanctions enforceable, including the micropayment rail [2] over which usage is settled and against which maintenance must be funded, are developed in the sections that follow. What the CPR lens fixes, here at the outset, is the shape of the problem: the graph degrades under uncoordinated use, and a graph that degrades needs governance, not just a price.

3 Why pure usage under-provides freshness

A route fee is a usage fee: it is paid by the agent that calls a route, in proportion to how often it calls. That is exactly the right instrument for *rationing* a route — whoever uses it more pays more — and exactly the wrong instrument for *funding the freshness of the graph as a whole*. The maintained graph is a non-rival good: my reading a fresh route does not consume it, and a thousand other agents can read the same fresh route at no marginal cost to me [5]. It is also close to non-excludable in the dimension that matters — once a route is verified fresh, the benefit of that freshness accrues to every agent that touches it, not only to the one whose fee happened to trigger the check. A good that is non-rival and hard to exclude is the textbook public good [5], and the textbook consequence follows: each agent prefers to let *someone else's* fee pay for the maintenance and then free-ride on the result. Sum that preference over a whole population of rational callers and maintenance is systematically under-supplied. Usage fees alone do not fail because the price is wrong; they fail because freshness is the wrong kind of good to price by the call.

The mirror image of the under-supplied public good is the over-supplied externality. When a maintainer lets a route rot — a schema drifts, an auth flow mutates, a parameter is silently renamed — the cost of that decay does not land on the maintainer who neglected it. It lands on every downstream agent whose call now fails, retries, or quietly returns wrong data. That is a negative externality in the precise Pigovian sense: a private action (neglect) whose social cost (aggregate failure across all callers) is borne by parties the actor never compensates [6]. Pigou's prescription for an uninternalised externality is not exhortation but a price on the harmful act, set so the actor faces the cost it imposes on others [6]. In a route graph the corresponding instrument is a bond the maintainer posts against its own freshness claim, slashable when the claim turns out false — the Pigovian corrective rendered as collateral rather than a tax. Staleness stops being a free option and becomes a liability the claimant actually carries.

These two failures point at the same structural conclusion, and it is the reason the security function cannot ride on the usage rail. The usage rail — stablecoin settlement bound to each `execute` over x402 [2] — is built to meter and route payment for calls that happen; it is excellent at exactly that and should stay narrow. But under-provision of a public good and the pricing of a negative externality are both *accountability* problems, not metering problems: they need an asset that can be bonded, staked, and slashed against the integrity of the graph itself, whose value is tied to that integrity rather than to call volume. That is the job we reserve for FDRY, kept strictly separate from the USDC usage rail for the same reason one does not pay a deductible out of the till: the thing that funds maintenance and the thing that prices its failure must be the asset whose worth rises and falls with the network being worth trusting at all. [proposed] — the bond-and-slash corrective is the sequenced security layer, not a shipped mechanism; the usage rail it sits beside is live [2].

4 Trust tiers

Not all traffic needs the same assurance, and pretending it does is its own kind of dishonesty — it taxes a price-check that can fail harmlessly at the same rate as a funds-moving call that cannot. So the graph is stratified into tiers. *Open* routes serve low-risk, idempotent, read-mostly traffic: a public listing, a schema lookup, a price quote, anything where a stale or wrong answer is cheap to detect and cheaper to retry. These carry no bond and make no guarantee beyond best effort; they are the fast, free majority of the graph and are honest about being best-effort. *Higher-trust* routes sit above them for authenticated, high-value, or side-effecting paths — moving money, mutating an account, anything where a silent failure is expensive and a wrong route is dangerous. The escalation is the point: an agent pays for assurance only on the calls that actually warrant it, and the open tier is never asked to pretend it is something it is not.

The load-bearing rule across all tiers is what *ranks* a route, and the answer is route quality — observed freshness, success rate, faithful replay — not the size of anyone's stake. We are emphatic about this because it is the exact line where a trust network degrades into a pay-to-win one: bonding buys a maintainer *eligibility* to make a slashable claim on a high-trust route, and it buys nothing further [1]. It does not buy placement, it does not buy a rank, and it does not let a well-capitalised maintainer outbid a better one for the top of a shortlist. The moment stake purchases ranking, the graph is selling position rather than reporting quality, and a quality signal you can buy is not a quality signal. Capital gates entry to the bonded tier; merit orders within it.

A trust tier needs a way for a stranger to evaluate a maintainer it has no prior reason to trust, and we reuse the ERC-8004 "Trustless Agents" model rather than inventing one [10]. It defines three registries that map cleanly onto a route maintainer: a portable **Identity** that follows the maintainer across contexts, a **Reputation** record of signed feedback that accrues from real outcomes, and a **Validation** surface where an independent party re-executes or otherwise checks a claim before trusting it. An Unbrowse maintainer is precisely one of these trustless agents — an identity that can carry reputation and be verified by someone who does not yet trust it — which is what lets the higher tier be evaluated on portable, checkable history rather than on assertion. [proposed] — the tier structure and its ERC-8004 binding are a design target with cited foundations [10], not a shipped registry; what ships today is the open graph and its usage settlement, with the bonded tier sequenced behind it.

5 Bonded, slashable maintenance

The trust tiers of the previous section raise an obvious question: what makes a higher-trust claim actually trustworthy? A route's freshness is not a fact the network can read off a register — it is an assertion someone makes about the world ("this endpoint still works, its schema has not drifted, its auth flow is unchanged"), and an assertion is only as good as what the asserter has to lose if it is wrong. So a maintainer who wishes to claim a route is fresh and safe at a trust tier *bonds* FDRY against that claim: the bond is collateral posted into the open, escrowed against the specific route attestation. A claim that later proves false, or that conflicts with another bonded claim about the same route, is *slashable* — the maintainer forfeits stake. This is collateralised accountability: the maintainer is paid to be right and made to pay for being wrong, so claiming freshness stops being free talk and becomes a position with skin in it.

The shape of this is not novel and we do not claim it as such; it is exactly the *accountable-safety* discipline of proof-of-stake finality. Casper the Friendly Finality Gadget [8] secures a chain not by trusting validators but by defining slashing conditions: a validator who signs two conflicting checkpoints, or otherwise violates the protocol's safety rule, can be proven to have done so and has its deposit destroyed. A maintainer who signs "this route is fresh" over a route that is in fact stale — or who signs two contradictory freshness claims — is in precisely the position of a validator violating a slashing condition: the misbehaviour is attributable to a specific bonded identity, and the penalty falls on that identity's stake rather than on the network at large. The accountability is local, provable, and costly, which is the only combination under which a stranger is rational to rely on a claim made by someone they have never met.

Finalising a trust-tier claim is not the act of one confident wallet. A single bonded maintainer asserting freshness is a single witness, and a single witness is not corroboration. The settlement of a tier claim is therefore a threshold event: FROST [9], a flexible round-optimised Schnorr threshold signature scheme, lets a quorum produce one valid signature only when t of n independent signers participate, with no single party able to forge the result alone. A trust-tier freshness claim is finalised by such a t-of-n quorum, so a tier does not advance on one operator's say-so and one dishonest or compromised maintainer cannot settle a claim by itself. The quorum is the structural form of the two-witness rule applied to finality.

We are deliberately precise about what the bond purchases. Bonding buys *credibility* — the right to make a claim a counterparty has reason to believe — and *eligibility* to participate at a trust tier. It does *not* buy ranking. The moment stake purchases placement in results, the search surface becomes pay-to-win and the asset securing trust degrades into the asset buying attention; the two roles are incompatible and we keep them apart by construction. Ranking is grounded in route quality (Section on trust tiers); the bond is a security deposit against a freshness assertion, nothing more. This entire layer — the bond, the slashing conditions, the threshold finalisation — is sequenced as future work [proposed]; the open and quality-ranked tiers do not depend on it, and we will not present a designed mechanism as a shipped one.

6 Challenge and dispute

A bond gives a false freshness claim something to lose, but it does not by itself decide *whether* a claim is false — a slashing condition is only as useful as the procedure that proves it was met. We resolve this the way the rest of the stack resolves trust: not by an operator's adjudication, but by independent corroboration. A challenger who believes a bonded freshness claim is wrong opens a dispute by staking against it. The challenger's stake is itself collateral, so a dispute is not a free heckle: a frivolous challenge loses its stake to the maintainer, and a sound one is rewarded from the slashed bond. Both sides now have skin in the same question, which is the precondition for the question being answered honestly rather than loudly.

Resolution is by independent re-execution. The disputed claim is a claim about an observable fact — does this route still return the expected result? — so it is settled by re-running the route from parties who do not trust either side and comparing what they observe. This is the two-witness principle made operational: a claim stands only on independent corroboration, never on one party's assertion (neither the maintainer's original claim nor the challenger's accusation is self-validating). ERC-8004's Validation registry [10] is the surface for exactly this — it defines independent re-execution, and where applicable ZK or TEE checks, as the mechanism by which one agent's work is verified by others who have no reason to favour it — so a freshness dispute is adjudicated by the re-execution evidence, not by a trusted referee.

For the dispute history to be auditable, the record of claims and challenges must itself be tamper-evident rather than a log a single operator could quietly rewrite. We therefore keep the sequence of freshness claims, challenges, and resolutions in an append-only, hash-chained log in the Certificate Transparency tradition [11]: each entry commits to its predecessor, so the history and its ordering cannot be altered after the fact without breaking every subsequent root, and any independent party — not just the platform — can verify that the log is append-only and consistent. The point of the CT analogy is precisely that correctness does not rest on trusting the operator: a misissued or back-dated claim is detectable by anyone auditing the log, exactly as a misissued certificate is detectable in CT. The challenge mechanism supplies the economic incentive to look, and the transparent log makes what they find undeniable. This challenge-and-dispute layer is sequenced together with bonding as future work [proposed]; it is the accountability complement to the bond, and like the bond it is a designed mechanism with cited foundations, not a shipped feature.

7 Attribution and the Sybil limit

A maintenance economy that pays for contribution has to answer a sharper question than "who touched this route?" — it has to answer "how much better is the route *because* you touched it?" We answer it with delta-based attribution: a contributor is rewarded in proportion to the marginal improvement their work makes to a route, not for mere presence on it. Tightening a schema so that fewer calls fail, adding an endpoint that was missing, repairing an auth flow that had drifted — each is scored by the difference it makes to the route's measured quality, so the reward tracks the increment, and a no-op edit on top of someone else's work earns nothing. This keeps the incentive pointed at real maintenance rather than at land-grabbing a popular route and collecting rent on another party's labour.

Delta-based attribution prices contribution honestly, but it does not, by itself, solve who is making the contribution — and here we have to be plain rather than reassuring. The marginal-contribution score is computed per identity, and in an open system identities are cheap. Douceur's result is the one that governs this [7]: without a central authority to certify that distinct identities correspond to distinct entities, a single entity can always forge many identities, so Sybil attacks are, in the general case, essentially always possible. An adversary who can mint identities for free can in principle split one contribution into many, or stand up a crowd of identities to manufacture the appearance of independent improvement. Bonding is our mitigation: requiring stake to participate in the attributed tier raises the *cost* of each forged identity, and graduated stake makes a large Sybil cohort expensive rather than free. But raising a cost is not closing a gap. Bonding makes the attack more expensive; it does not formally guarantee that distinct attributed identities are distinct entities, which is exactly the guarantee Douceur shows cannot be had without a trusted certifier.

We therefore label this an acknowledged open problem rather than a solved one. [proposed] Formal Sybil resistance for the attribution layer is future work with a cited foundation [7], not a property we claim today; the bonded tier narrows the economically rational attack surface, and the honest statement is that it narrows it, not that it eliminates it.

8 The token: why FDRY, and why fair-launched

This section makes one argument, and it is an argument about security, not about markets. The conclusion is uncomfortable for a token paper to state plainly: *a coordination asset launched for money cannot become trustworthy, so the only way to make it trustworthy was to launch it for something else and hide it from the money-motive at the root.* That is not a marketing posture dressed as virtue — it is a property the system needs in order to function, and the rest of this section derives it.

One asset, one job. The first discipline is to refuse to conflate two surfaces that the rest of crypto routinely fuses. Usage — a paid `execute`, a settled call — does not need a token at all; it settles in stable USDC over x402 [2], and we are emphatic that it stays that way. FDRY exists for exactly one job the stablecoin cannot do: the security/accountability layer, where a maintainer's bond, a challenger's stake, and a slashing penalty must be denominated in an asset whose value is tied to the network's own integrity [1]. Payment surface and security stake are kept strictly separate, because the moment a holder is forced to spend the security asset to *use* the network, the asset becomes a tollgate and the holder a beggar-at-the-gate. This is the old wisdom that "No one can serve two masters" (Matt 6:24) read as an engineering constraint: [shipped] USDC settles usage; FDRY only bonds. An asset asked to do both jobs does neither honestly.

Why the distribution is a security parameter. The non-obvious constraint is on FDRY's *distribution*, and it falls straight out of the same witness logic that governs route trust. Accountability is credible only when many *independent* parties hold the bond — the same way a claim stands on two independent witnesses rather than one. A security asset concentrated in insider hands is a single witness wearing many hats: it contradicts the very decentralisation it is meant to underwrite. So FDRY launched fair by necessity, not branding: no pre-sale, no team allocation, no insider round; supply placed on a public Solana pool at launch, acquirable only on the open market at the same price for the founder as for everyone else. The fair launch is therefore not a gesture of goodwill — it is the parameter that makes the bond mean anything. [shipped] (fair launch; live platform-fee collection). The vault cycle that returns the platform's USDC fee to the distributed holders — staking by abiding rather than by lock-up — is [proposed].

Why money could not be the point. There is a deeper reason the distribution had to be fair, and it is structural rather than financial: it concerns the order in which trust and value can be created. A network that puts money first inverts its own foundation. As we argue in the descent paper [1], if the token is launched to enrich its issuer, then the issuer's incentive at the root is extraction, and every participant downstream correctly discounts the system's trust claims accordingly — the rot is at the origin, so no amount of mechanism layered above it recovers credibility. An accountability layer whose entire job is to let strangers trust each other's claims *cannot* sit on a money-first root, because a root optimised for the issuer's gain is precisely the thing strangers are right to distrust. FDRY clears this bar only because money was not its point: it was distributed with no issuer advantage and then repurposed as the security stake. The asset earns trust the same way a route does in this system — by what it does for the network, corroborated by independent holders, not by what it promised its founder. This is the grain-of-wheat constraint stated as economics: a seed kept for itself "remains by itself alone," and bears only by falling into the ground first (John 12:24). Trust precedes value here; the value is the fruit of the trust, never its bait.

Disclosed in full, including the address. None of this is a license to be coy. A trust paper that is evasive about money is not trustworthy, so we state the economics plainly — who pays, who earns, who opts in — and we print the contract address on purpose: `2ZiSPGncrkwWa6GBZB4EDtsfq7HEWwkwsPFzEXieXjNL`. What the doctrine refuses is the *investment narrative*, the framing of the token as a bet on its own price. The "what" is fully disclosed; only the money-as-motive framing is denied, because that framing is exactly what would poison the soil the security layer has to grow in. We make no promise of profit: the mechanism is the claim, not a guarantee.

9 What is built versus proposed (no fabricated green)

In the spirit of the companion paper [1] — and of not selling a roadmap as a changelog — we separate what runs today from what is designed but unbuilt:

- [shipped] The live three-way split: every paid execution settles to the indexer, the domain owner, and the platform in one transaction over x402 [2].

- [shipped] Fair-launched FDRY: supply placed on a public pool at launch with no pre-sale, team allocation, or insider round.
- [shipped] Platform-fee collection on settled executions.
- [shipped] The signed, append-only JSONL route ledger — the local, hash-chained record of route claims.
- [proposed] Bonding, challenge, and slashing: the collateralised-accountability layer in which a maintainer's stake is forfeit on a false freshness claim.
- [proposed] On-chain Merkle-root checkpoints of the ledger, batching many signed entries into a single transparency-style root [11].
- [proposed] Formal Sybil resistance for attributed contribution, which remains an open problem with a cited foundation [7].

Treat the [proposed] list as a research agenda with cited foundations, not a feature list with a release date. The gap between those two lists is the entire difference between a whitepaper and a press release, and we intend to stay on the correct side of it.

10 Conclusion

The thesis of this paper compresses to one sentence: the graph is the asset. Every site an agent automates today is re-discovered from scratch, per agent, per call — the browser tax paid over and over. Unbrowse turns that one-off robotic-process work into a shared, maintained route graph: discover once, reuse forever, route the cost through one layer instead of re-paying it everywhere [1]. The crucial asymmetry is that *any single route is copyable, but the maintained graph is not.* A route can be lifted; what cannot be lifted is the standing work of keeping thousands of routes fresh as the web shifts under them, attributed to who captured them, and safe enough that another agent will stake its task on them.

That is why maintenance, not discovery, is the economic engine. The value compounds in the upkeep: realised payment over x402 [2] funds the substrate that keeps the graph trustworthy, and the network pays whoever actually created the value of a reused route — the indexer who captured it, the domain owner whose service is called, and the platform that secures the settlement. Each share is the dominant strategy for the role it pays, so the graph's coverage grows, cooperation out-competes blocking, and the platform earns only when a settlement it secured actually happens. FDRY sits underneath all of it as the bonded, fairly-distributed security stake — the asset that makes "trust this route" a claim with collateral behind it, not a hope.

We have told this as the *what* throughout, and stopped exactly where the *how* begins. The primitives that make route discovery cheap and the graph fresh are closed by design [3], for the abuse reasons stated there — the boundary is not secrecy for its own sake but the difference between giving agents real APIs and handing out an unattributed scraping fleet. Internal APIs were a great first layer. The maintained, attributed, bonded graph above them — kept fresh, paid for in proportion to the value it creates, and underwritten by an asset that had to be hidden from money before it could be trusted with it — is the layer that was never all you needed, and is the one worth building.

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